Quick guide: run the simulation

The baseline physics example does not need the optional AI model

1. In a terminal, go to the project folder:

```
/Users/danielsinambela/Documents/physics-aware-reservoir-agent
```

2. Run the baseline example:

```
.venv/bin/python examples/run_baseline.py
```

The program prints the flow rate and physics checks, then saves the plot to outputs/baseline.png. In the run described in the report, all five checks passed.

Other useful commands

```
.venv/bin/python examples/validate_physics.py
```

Test several physics cases.

```
.venv/bin/python examples/heterogeneous_reservoir.py
```

Run a less uniform rock example.

```
.venv/bin/python -m pytest tests/ -v
```

Run the automated test suite.

If Python reports a missing package, install the project dependencies:

```
.venv/bin/python -m pip install -e .
```

A virtual environment is a project-specific place for Python packages. The .venv/bin/python command uses it.